

## ABOUT EnPoWER

EnPoWER, which stands for the Engineering Oriented Promotion of Work Experience and Research, serves as the undergraduate research promotion arm of the institution. Our primary objective is to facilitate greater accessibility to **research core-related and industry opportunities** for undergraduate students.

EnPoWER is dedicated to enhancing the involvement of undergraduate students from IIT Bombay in the research and development of engineering-based solutions for industry challenges. This is achieved through opportunities, recognition, and heightened awareness. Through events such as "Enthuse," "SURP," and "ResCon," EnPoWER acts as a bridge, connecting students with professionals from the scientific community, including professors, seniors, and industry experts.

## WORK DONE IN THE TENURE

- **RESCON 2024 (13th and 14th January, 2024)**

EnPoWER, supported by IRCC, hosted the second iteration of ResCon 2024, a research conference for undergraduate students held at IIT Bombay. The primary objective of the event was to ignite curiosity in engineering undergraduates and provide a platform for them to showcase their research projects. In addition to over 500 students from IIT Bombay, more than 200 students from engineering institutions across India participated in ResCon 2024.

The theme of ResCon centered around 'Soch Vichar Achar', where "Soch" symbolized inquisitive thinking, encouraging students to delve into diverse subjects with a curious and questioning mindset. "Vichar" underscored the significance of inquiry as an integral part of the research journey, urging students to delve deeply and critically into their chosen research domains, challenging conventional beliefs and assumptions. Through the lenses of "Soch" and "Vichar", the aim was to prompt students to explore the pivotal role of curiosity in research, advocate for interdisciplinary collaboration, and nurture creativity and innovation. "Achar" represented the highlight of ResCon 2023, the Research Presentation Competition.

EnPoWER organized a series of competitions, workshops, and discussions over two days to immerse participants in the realm of research. In addition, students, professors, industry professionals, and research scholars also contributed to the success of ResCon 2024. IRCC collaborated closely throughout the conceptualization and execution phases of ResCon 2024.

**Day 1: 13th January, 2024**

## **Opening Ceremony**

**Keynote Speaker - Dr. Vijay Kumar Saraswat**

**Topic: Bridging the Divide between Industry and Academia**

At the opening ceremony held in LTPCSA, we were honored to have Prof. Sachin Patwardhan (Dean R&D, IIT Bombay) deliver the opening speech. He highlighted the various research initiatives underway at IIT Bombay, discussed the central facilities, and encouraged attendees to engage in research. Dr. V.K. Saraswat, a distinguished Padma Bhushan awardee and former Director General of the Defence Research and Development Organisation, as well as former Chief Scientific Advisor to the Indian Minister of Defence and current member of NITI Aayog, graced us as the chief guest. He presented an in-depth talk on bridging the gap between industry and academia, a topic highly pertinent to the conference. His presentation addressed the main challenges, effective strategies, and key stakeholders involved in narrowing this divide.

## **Achar Research Presentation Competition**

The highlight event at ResCon 2024 was the research presentation competition "Achar," providing undergraduate students with an opportunity to showcase their research projects to their peers and experts. The competition comprised three rounds, structured as follows:

### **Abstract Submission Round**

For the initial elimination phase, students or teams were required to submit abstracts outlining their research work, emphasizing the progress made, novelty, and relevance to the industry. We received 103 abstract submissions spanning the following domains:

- Concrete Technology (2)
- Aerospace and Structural Engineering (3)
- Astronomy (3)
- Biology, Healthcare or Health Sciences (10)
- Blockchain Technology (5)
- Chemistry and Materials Science (15)
- Civil Engineering (2)
- Computer Science or Artificial Intelligence / Machine Learning (32)
- Education (2)
- Electrical and Electronics (5)

- Environmental or Agricultural Sciences (4)
- Mathematics and Statistics (1)
- Metallurgy and Materials (2)
- Robotics and Drones (2)
- Social Sciences (Sociology, Psychology, Economics, etc.) (2)
- Theoretical Physics and Mathematics (5)
- Thermal Science and Fluid Mechanics (8)

Following an objective scoring process conducted by the EnPoWER team, the top 40 submissions were selected to showcase their research work through a poster presentation round held on day one of the conference.

## Poster Presentation Round

In this round, the 40 shortlisted teams or individuals showcased their research work. Each team presented their poster for a duration of 120 minutes. The top scorers, highlighted in green, proceeded to the 15-minute presentation round based on the following judging criteria: 80% of the final score was determined by professors' evaluations, while the remaining 20% were through audience votes through a coupon voting system. The panel consisted of 4 judges from various domains which included:-

- Prof. Prashant P. Date - Mechanical Engineering
- Prof. Chandramouli Subramaniam - Chemistry
- Prof. Aditi Chaubal - Economics
- Prof. Puja Padhi - Economics

Judging criteria:

- Content and Depth of Research (15 marks): Well-structured and researched work, put forward with clarity of thought.
- Presentation Skills (15 marks): The quality of the presentation and their ability to communicate and defend their idea.
- Layout/Design of Poster (5 marks): The overall arrangement and visual appeal of the poster.
- Impact and Relevance (5 marks): The importance and potential of the idea presented in the real world.

**Conveying Complexity with Prof. Arnab Dutta**

In an engaging connecting event, Prof. Arnab Dutta from the Chemistry Department shared his research journey and experiences. He eloquently described a day in the life of a researcher, highlighting its rewards. He emphasized the ideal attitude a scientist should possess and discussed the common challenges faced in projects and how to address them. His talk was thoughtfully crafted and served as a motivating session for all.

### **Fireside Chat With Dr. Rohit Hippalgaonkar**

We had the privilege of hosting Dr. Rohit Hippalgaonkar, the Associate Director at OLA Electric for a fireside chat, who graciously shared insights into his remarkable journey from IIT Bombay to OLA Electric. With a solid foundation in mechanical engineering from IIT Bombay, followed by advanced studies at Cornell and Purdue University. Transitioning from esteemed companies like Ford Motors to his current role at OLA Electric, Rohit shed light on the practical applicability of academic concepts learned at IIT Bombay, emphasizing the importance of bridging the gap between theory and real-world experience. He also reflected on the unexpected challenges encountered in the industry and the resilience required to overcome them. Additionally, Dr. Rohit shared the experience of a typical day in the engineering field, highlighting the fulfillment derived from solving complex problems and driving innovation. His journey is an example of the significance of perseverance, passion, and continuous learning in achieving professional excellence.

### **Ethical Dilemmas in Research (Debate)**

This event featured a formal debate focusing on ethical dilemmas prevalent in research, with a specific emphasis on the contentious topic of testing on animals. The debate comprised four members from IIT Bombay Debate Society - Nayantara R, Mahor R, Harshvardhan J, and Harshith R, each representing diverse perspectives on the ethical considerations surrounding this issue. Throughout the debate, participants engaged in articulate debate, presenting compelling arguments and counterarguments regarding the use of animals in research. Key topics discussed included the ethical implications of animal testing, the potential benefits and drawbacks of such research practices, and alternative methods for conducting scientific experiments. The debate sparked thought-provoking discussions about the ethical responsibilities of researchers, the moral implications of using animals in research, and the need for ethical guidelines to govern such practices.

## Day 2: 14th January, 2024

### **Women in STEM**

The Women in STEM panel discussion held on Day 2 of ResCon 2024 at IIT Bombay was an insightful and inspiring session, featuring accomplished women scientists sharing their journeys, challenges, and insights with the audience. The panel consisted of:

- Dr. Sujatha Ramdorai (Mathematician) - Ramanujan Award and Padma Bhushan Winner
- Prof. G.C. Anupama (Astronomer) - First Woman President of the Astronomical Society of India
- Dr. Sandhya Koushika (Neuroscientist) - IECS Awardee, Women's Cell at TIFR
- Prof. Sumati Surya (Theoretical Physicist) - First Woman President of IAGRG
- Dr. Swati Patanka (Bioscientist) - Leading Gender Initiatives at IIT Bombay

The panel discussion involved multiple segments some of which are as follows:

- Career Beginnings and Mentorship: The panelists shared their early experiences in research and highlighted the role of mentors, especially female mentors, in shaping their careers.
- Challenges Faced by Researchers: Dr. Sujatha Ramdorai shed light on key challenges faced by researchers in India, fostering a deeper understanding of the research landscape.
- Memorable Career Moments: Each panelist shared a memorable moment from their career that significantly impacted their outlook as a researcher, showcasing the personal and professional growth in their journeys.
- Equitable Representation in STEM: The discussion explored strategies to ensure more diverse voices and equitable distribution of power in STEM fields, emphasizing inclusivity and opportunities for all.
- Favourite Projects and Future Technologies: Panelists discussed their favorite projects, simplified for the audience, and highlighted upcoming technologies in their respective fields that they are excited about.
- Rapid Fire Round: A fun and engaging rapid-fire round allowed the audience to learn more about the panelists' experiences, preferences, and perspectives on research life and beyond.
- Audience Questions: The session also included insightful audience questions on topics ranging from advice for aspiring researchers, the role of mathematics in daily life,

public awareness about global health issues like malaria, communicating complex theoretical physics concepts to the public, diversity in academia, the intersection of AI/ML with neuroscience, and more.

Overall, the panel discussion was a resounding success, leaving the audience inspired and motivated to contribute meaningfully to the world of science and technology. We extend our gratitude to the panelists, moderators, and attendees for their valuable contributions to this enriching event.

### **Re-Decrypt - Research Paper Crypt Hunt**

The Re-Decrypt competition was another highlight of Day 2. The event brought together teams of two participants who engaged in decrypting questions related to a curated collection of 20 research papers spanning various fields of study. The competition format involved utilizing online research tools and libraries to unravel intricate details such as references, ISSN numbers, graphs, authors, and more, showcasing participants' analytical and research skills. With cash prizes worth INR 30,000 at stake, including amounts of ₹12,500, ₹10,000, and ₹7,500 for the top three winning teams, the competition added excitement and motivation to the research-focused atmosphere of the conclave. Winners were announced during the closing ceremony, adding a rewarding conclusion to this intellectually stimulating competition.

Re-Decrypt was a successful event that enriched the research atmosphere of the conclave. It prompted participants to explore research literature, enhance analytical skills, and foster teamwork. The competition's challenging format made it a memorable and rewarding experience for all.

### **Achar Finals - 15-Minute Presentation**

The top 10 teams from the poster presentation rounds presented a detailed description of their project work in front of the judging panel. This included their motivation, introduction, objectives, methodology, results, conclusion, and application. The panel consisted of 6 judges from various domains which included:-

- Prof. Dipanshu Bansal - Mechanical Engineering
- Prof. V. Jothiprakash - Civil Engineering
- Prof. Najeeb Shariff - Civil Engineering
- Prof. E. Chandrasekhar - Earth Sciences
- Prof. Sandip Kar - Chemistry
- Prof. Soumya Bera - Physics

The judging criteria for the finals were the following:

- Need and Impact (10 points): The importance and potential of the idea presented. If the alignment with the present industry standards was adequately addressed.
- Content and Clarity (20 points): Well-structured and researched work, put forward with clarity of thought. The participant(s) can respond to inquiries posed by the panelists.
- Presentation Skills and Preparedness (20 points): The quality of the presentation and the ability to communicate and defend the idea.

The 15-minute presentation consisted of an initial 10 minutes for the participants to present their work followed by a 5-minute Q&A session where each team had to answer approximately 4-5 questions from the judging panel. All the finalists were awarded cash prizes, with the following split:

| <i>Position</i> | <i>Amount</i> |
|-----------------|---------------|
| Winner          | ₹65,000       |
| 1st Runner-Up   | ₹35,000       |
| 2nd Runner-Up   | ₹20,000       |
| 4th             | ₹8,500        |
| 5th             | ₹6,500        |
| 6th             | ₹5,000        |
| 7th             | ₹4,000        |
| 8th             | ₹3,000        |
| 9th             | ₹2,000        |
| 10th            | ₹1,000        |



## **Keynote Speaker - Dr. Anil Kakodkar**

### **Topic: Revolution & Future of Research**

Padma Vibhushan Awardee, Dr Anil Kakodkar was the former chairman of the Atomic Energy Commission of India and former director of the Bhabha Atomic Research Centre. He spoke about Revolutions in Research and the Future of Research, in the context of India and how that can bring about development that can benefit society as a whole.

He began by speaking about India and where it stands in the world as technology is bringing about change and society is shifting from focusing on industry to the knowledge era. He then spoke about the importance of higher education and universities and how they can be utilized to create both knowledge and value in our country. He also spoke on reducing disparity and bringing technology, resources, and quality education from schooling to higher education to villages. Finally, he spoke on the importance of doing quality research that is integrated with society and the economy to promote equality and overall social welfare.

## **Closing Ceremony**

Finally, to conclude ResCon 2024, we had the Achar award distribution ceremony. Dr. Anil Kakodkar presented the award to the finalists who were invited to the stage and rewarded with trophies and certificates. This was followed by a vote of thanks delivered by the ISAA and Head, EnPoWER.

- **CRAFT YOUR RESEARCH PERSONA (10th January 2024)**

We conducted a session on academic writing for researchers and were joined by 3 accomplished speakers who had significant experience on writing academic papers.

First, we had Prof. Rekha Santhanam from the Mathematics department at IIT Bombay who broke down the essence of writing academic papers. She spoke about the art of communicating through research papers and a bit about her personal experience on academic writing.

She was followed by Pranava Singhal, a student from the Electrical Engineering department at IIT Bombay. He provided an introduction consisting of why you should consider pursuing research, how you can get started, and how the journey looks from start to end. Coming to writing, he described the anatomy of a paper and gave tips on how to find research papers and effectively read research papers. He also spoke about external research opportunities

Finally, we had Ankan Mukherjee who is pursuing a PhD from Caltech join us virtually. He spoke on how to get started on writing a paper. He spoke about the template of a paper and



then on proofreading and professor-student interaction. Finally, he spoke on what happens after you write a paper and send it to a journal.

After each speaker, students were allowed to ask them questions about their work and academic writing.

- **SUMMER INTERNSHIP AT IRCC**

EnPoWER, in collaboration with the Industrial Research Consultancy Centre (IRCC) at IIT Bombay, announced the selection of candidates for a summer internship opportunity. The internship program, which will span 2 months during the summer break, offers selected candidates room retention as needed and a monthly stipend of INR 15,000. Interns were to work closely with IRCC to curate a compendium of patents, technologies, and research facilities at IIT Bombay, aimed at attracting industrial investments and providing valuable hands-on experience to students.

- **WURP 2023 (Winter Undergraduate Research Program) (October 2023 - present)**

*In coordination with the Department Research Coordinators*

EnPoWER brought the first-ever edition of the Winter Undergraduate Research Programme 2023. The Winter Undergraduate Research Program of 3 sub-programs:

- 1. iSURP 2023 (In-Semester Undergraduate Research Program):**

Similar to SURP, the In-Semester Undergraduate Research Program (iSURP) is a program that facilitates the process of students taking up projects under the professors during the winter break and these projects will continue into the Spring Semester 2024.

- 2. CORE 2023 (Curriculum Oriented Research Experience):**

CORE stands for Curriculum Oriented Research Experience and the main objective of the program is to provide an opportunity to second and third-year undergraduate students to apply the knowledge they gain from academic courses (from the current Autumn Semester) to research and real-life industry problems. The vision behind CORE is to encourage students to give research a chance and while doing so, not get demotivated by lack of understanding/knowledge of higher-level concepts. These projects will also continue into the Spring Semester 2024.

An overview:

- Revamped the WURP portal for students to take up projects to make it user-friendly

- Received a total of 68 projects with 56 for iSURP, 11 for CORE, and 1 course for LabVentures from professors.
- Procured projects from non-UG departments like BSBE, SysCon, CTARA, Educational Technology, SOM, CUSE, Climate Studies and IEOR.
- Statistical data for WURP 2022 is as follows:

| Department                              | Number of iSURP projects | Number of CORE projects        |
|-----------------------------------------|--------------------------|--------------------------------|
| Aerospace Engineering                   | 1                        | 1 (AE223)                      |
| BSBE                                    | 3                        | -                              |
| C-MInDS and SysCon                      | 1                        | -                              |
| Centre for Policy Studies               | 1                        | -                              |
| Chemical Engineering                    | 4                        | -                              |
| Chemistry                               | 2                        | -                              |
| Civil Engineering                       | 4                        | -                              |
| Climate Studies                         | 1                        | -                              |
| Computer Science Engineering            | 3                        | 2 (CS6001)                     |
| CSRE                                    | 1                        | -                              |
| Earth Sciences                          | 2                        | -                              |
| Educational Technology                  | 1                        | -                              |
| Electrical Engineering                  | 1                        | 3 (EE229, EE678, EE338, EE501) |
| Energy                                  | 3                        | -                              |
| Environmental                           | 2                        | -                              |
| Mechanical Engineering                  | 9                        | 3 (ME221, ME346, ME800)        |
| Metallurgy and Material Sciences (MEMS) | 1                        | 2 (MM305)                      |
| Multidisciplinary                       | 2                        | -                              |

|                            |    |    |
|----------------------------|----|----|
| Physics                    | 13 | -  |
| School of Management (SOM) | 1  | -  |
| TOTAL                      | 56 | 11 |

- SURP 2023 (Summer Undergraduate Research Program) (June 2023 -present)**

*In coordination with the Department Research Coordinators*

SURP is a program that facilitates the process of students taking up projects under the professors during the summer as well as the professors finding it easy to float projects to applicants from interdisciplinary fields. (SURP 2023 was conducted with both offline as well as remote-online projects)

An overview:

- A SURP portal was designed by EnPoWER and the Web Team to display the projects
- A total of 79 projects were floated by 65 faculty members of IIT Bombay
- Each applicant was eligible to apply for 2 projects; in total, 233 applications were received
- Projects from professors from 21 departments and centers of excellence such as the IEOR, SysCon, Climate Studies, and Educational Technology
- The form was re-floated for projects with zero applicants, and selected candidates were allocated projects accordingly
- *Progress Dashboards* were introduced this year, taking weekly updates resulting in active engagement of the candidates with their projects
- *The Mid-Term Report* submissions were mandated to continue with the project. An impressive 63 submitted the report
- Statistics of SURP 2023 projects are as follows:

| Department | No. of Projects Floated |
|------------|-------------------------|
| BSBE       | 1                       |
| C-MInDS    | 2                       |
| Chemical   | 12                      |

|                                |    |
|--------------------------------|----|
| Chemistry                      | 3  |
| Civil                          | 5  |
| Climate Studies                | 3  |
| Computer Science Engineering   | 1  |
| CTARA                          | 1  |
| Earth Sciences                 | 1  |
| Educational Technology         | 3  |
| Electrical                     | 12 |
| Energy Science and Engineering | 4  |
| Environmental Sciences         | 2  |
| IDC                            | 1  |
| Mathematics                    | 1  |
| Mechanical                     | 8  |
| MEMS                           | 3  |
| Physics                        | 3  |
| SOM                            | 4  |
| SysCon                         | 5  |
| Inter-disciplinary             | 4  |
| TOTAL                          | 79 |

- **ENTHUSE 2023 (2nd August 2023)**

*In coordination with the IRCC*

More than 1400 first-year undergraduate students from IIT Bombay participated in high spirits, attending ENTHUSE 2023 which took place simultaneously across three venues: VMCC, F.C. Kohli, and LT-PCSA. The allocation of students to venues was based on their respective departments. This arrangement allowed for a diverse representation of

departments, with eighteen professors and three Deans, each hailing from distinct departments, sharing their research work and offering insights into what newcomers can expect from research opportunities at IIT Bombay. The presentations maintained an informal tone while being captivating and rich with information. Refreshments from Dhun were arranged and served to the attendees before the commencement of the event.

Furthermore, this year's edition introduced some captivating activities and quizzes, including Instagram challenges and trivia quizzes, all designed to enthrall the audience. Enthuse T-shirts were gifted to the winners of the interactive games and to the professors. Professors were also acknowledged with plants, as a small token of appreciation.

The event kicked off with a welcoming speech from IRCC, followed by EnPoWER. EnPoWER also talked about the different research opportunities for undergraduates at the institute. Every effort was made to make the event interesting and interactive by having quizzes and activities about research, IIT Bombay, and important scientific work in between. This was done to get the people at the event to talk and engage with each other. Professors from IIT Bombay gave talks to encourage students to think about doing research in different departments. Both students and professors really liked the event. Students were intrigued by new research opportunities from KCDH, SOM, DESE, and CTARA. Professors told EnPoWER after the event that they thought it was a great way to inspire students to think about research and that they were excited for more events like Enthuse. We floated an event feedback form among the freshman students. Based on the response, the way the event was organized, the interaction, and the hosting all got positive feedback from the audience.

The flow of the event was as follows: -

| VMCC B Nag Auditorium                                                                        | P.C. Saxena Auditorium                                                                       | F.C. Kohli Auditorium                                                                      |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <i>EnPoWER introduction +<br/>Introductory activity</i><br>Aarohi Deshpande<br>Reeyansh Shah | <i>EnPoWER introduction +<br/>Introductory activity</i><br>Sarvada Andhare<br>Sara Atnoorkar | <i>EnPoWER introduction +<br/>Introductory activity</i><br>Pranjal Deshmukh<br>Isha Laddha |
| <i>Prof In-Charge</i><br>Prof. Varun Bhalerao                                                | <i>Prof In-Charge</i><br>Prof. Arpita Mondal                                                 | <i>Prof In-Charge</i><br>Prof. Abhijit Majumder                                            |
| <i>Address by Previous Dean<br/>(R&amp;D)</i><br>Prof. Milind Atrey                          | <i>Address by Dean (R&amp;D)</i><br>Prof. Sachin Patwardhan                                  | <i>Address by A. Dean (R&amp;D)</i><br>Prof. Bhandarkar                                    |
| <i>Address By Professors:</i>                                                                |                                                                                              |                                                                                            |

|                                                             |                                                  |                                             |
|-------------------------------------------------------------|--------------------------------------------------|---------------------------------------------|
| <i>MEMS</i><br>Prof. Tanushree Choudhury                    | <i>Aerospace</i><br>Prof. Prabhu Ramachandran    | <i>Chemical</i><br>Prof. Guruswamy          |
| <i>DESE</i><br>Prof. Sagar Mitra                            | <i>Civil</i><br>Prof. Swagata                    | <i>Electrical</i><br>Prof. Kasturi Saha     |
| <i>IEOR</i><br>Prof. Jayendran Venkateswaran                | <i>Computer Science/KCDH</i><br>Prof. Amit Sethi | <i>Economics</i><br>Prof. Subrato Bannerjee |
| <i>Engineering Physics -</i><br>Prof. Suddhasatta Mahapatra | <i>ESED/Climate Studies</i><br>Prof. Srinidhi    | <i>IDC</i><br>Prof. Girish Dalvi            |
| <i>Mechanical</i><br>Prof. Arindrajit Chowdhury             | <i>DSSE -</i><br>Prof. Anuradha                  | <i>SYSCON</i><br>Prof. Leena Vachhani       |
| <i>SJMSOM</i><br>Prof. Piyush Pandey                        | <i>CTARA</i><br>Prof. Anand Rao                  | <i>BSBE</i><br>Prof. Roop Malik             |
| -                                                           | <i>CTARA</i><br>Prof. Bakul                      | -                                           |

## ● DEPARTMENT RESEARCH GROUPS (IIT Bombay Research Community)

To foster interaction between students interested in taking up research and to ignite a general engineering curiosity in them we formed the department research groups. The Department Research Community (DRC) initiative is focused on engaging third-, fourth-, and fifth-year students in dedicated research groups within the department. The process involved identifying 1-2 students from each year to help pinpoint individuals passionate about research. A key feature was creating exclusive groups and maintaining a closed structure to enhance the community's value. Membership was restricted to invitation or recommendation only. The initiative aims to kickstart with ice-breaker activities and initial events.

The vision behind the Department Research Groups is to create a platform where all interesting research can be discussed and any queries related to the future options in research are cleared within the community while the vision behind the sessions is to engage curious minds to become better engineers.

- **STUDENT RESEARCH @ TECH RnD EXPO (4th and 5th November 2023)**

*In coordination with the Institute Technical Council*

EnPoWER, in partnership with the Institute Tech Council, identified students who had previously participated in SURP or iSURP programs, as well as those who had completed individual projects, to share their research endeavors at the Tech R&D Expo. The primary goal of this undertaking was to present the research work of our fellow students to a diverse audience, including investors, professors, and individuals from both the city and the IITB student community. EnPoWER curated a selection of 12 undergraduate student research projects spanning different departments for this event.

- **CURIOUS COMMUNITY DISCUSSIONS**

To foster interaction between students interested in taking up research and to ignite a general engineering curiosity in them, we organized the Curious Community Discussions and established a discussion platform for students from all years and departments. The vision behind the Curious Community is to create a platform where all interesting research can be discussed and any queries related to the future options in research are cleared within the community while the vision behind the sessions is to engage curious minds to become better engineers. The following discussions have been undertaken so far:

- **Curious Community Session 1- Rumpled Papers Freshers' Challenge:** To inspire freshmen to take up research, we conducted a mini Research Paper writing competition. The students were allotted problem statements based on fictional books, series, and movies randomly and they were expected to come up with a plausible theory and present it in the format of a research paper. The session took place in LA202, starting from 9:30 P.M. with 80+ students participating. The students also enthusiastically volunteered to present their introductions there. Organizing such an event not only encourages research and critical thinking but also provides students with valuable skills and experiences. It can also be a fun way to engage with popular culture and connect it with academic pursuits.
- **Curious Community Session 2 - Conveying Complexity:** We organized the "Conveying Complexity" workshop at IIT Bombay, aimed at challenging 1st-year UGs with a series of intellectually stimulating games. With 35+ teams, each activity was designed to engage participants and filter out based on their creativity, problem-solving skills, and teamwork.



## Event Overview

The event featured four challenging activities that pushed participants to think innovatively and collaborate effectively.

- **Activity 1: Cup In An Island**

Participants were given materials like ropes, rubber bands, ice-cream sticks, bamboo sticks, scissors, tape, and glue. Their task was to design a device or strategy to remove a party cup from a small circle without touching it, using only their device. The challenge emphasized creativity and precision in engineering.

- **Activity 2: Zoom Puzzle**

The Zoom puzzle presented 30 jumbled pages with successive pictures, testing participants' ability to sort and arrange them in the correct order. This task required keen observation and logical thinking to decipher the sequence of images accurately.

- **Activity 3: The Tallest Marshmallow**

Teams were provided with sticks of spaghetti, tape, string, and a marshmallow. Their challenge was to build the tallest structure capable of sustaining the marshmallow on top. This activity tested participants' engineering skills, creativity in design, and structural stability.

- **Activity 4: Conveying Complexity Research Puzzle**

The final activity involved decoding and sequencing sections of research papers to unlock passwords and access complete papers. Teams had to choose a research paper title and correctly identify and sequence its abstract, introduction, methodology, and conclusion sections out of jumbled sections from 20 research papers. This task required deep understanding, critical thinking, and teamwork to generate a four-letter password that granted access to the complete research paper.

The winners were given Amazon vouchers.

- **NOVICE'S GUIDE (Released on 18th August 2023)**

Link to the booklet -  [Novice's Guide](#)

EnPoWER curated and released the first version of the Novice's Guide during the freshmen orientation season to make freshmen and sophomores aware of the research opportunities available at the undergraduate level. It aims to act as a one-stop guide for all research project-related queries and hence it includes a step-by-step guide to approach a professor for

a project and a general project experience.

It also includes information about the Undergraduate Research Award (URA), details about the application procedure, and eligibility criteria. Another highlight of the booklet is the invaluable experiences and advice from seniors who have won this prestigious award. Finally links to additional resources such as the Resume Repository and Core Intern Directory have been added to the booklet.

- **GRAD AND UNIVERSITY INTERNS SURVEY**

To gather feedback and insights from graduate students and universities regarding internship experiences, we recently conducted 2 surveys by floating Google Forms. It aimed to ascertain the channels and methods through which students secured admission into various academic institutions and fields of study so that they could help the next batch of juniors. Participants were asked to provide details regarding the means utilized to gain admission, including standardized tests, academic performance, extracurricular activities, recommendation letters, and personal statements. The survey sought to gain insights into the diverse pathways that students undertake to enter their chosen fields of study and institutions.

- **INTERNSHIP GUIDANCE PROGRAM**

*In coordination with Career Cell, UGAC*

1. **CORE INTERNSHIP BOOKLET (Released on 13th September 2023)**

Link to the booklet - [Core Internship Booklet](#)

EnPoWER, in collaboration with Career Cell, unveiled the second edition of the Core Intern Booklet during the internship season. The primary objective is to streamline the critical internship process and provide comprehensive assistance to students in their preparation endeavors. Serving as a singular point of reference, the booklet encompasses a robust database comprising 80+ core internship programs spanning various universities in India and abroad.

The booklet featured an extensive directory showcasing accomplished seniors who have successfully undertaken core internships in previous years, detailed insights into application procedures, eligibility criteria, and information on available scholarships for a diverse range of internship programs. The booklet also offers a valuable section dedicated to firsthand experiences and advice from seasoned seniors who have actively participated in core internship programs and pursued research initiatives.

Furthermore, the booklet integrates links to supplementary resources, such as the Resume Repository and Core Intern Directory, enhancing its utility as a comprehensive guide for aspiring interns.

## 2. UNIVERSITY INTERN SESSION (12th October 2023)

EnPoWER organized a University Intern Session aiming to aid the preparation of third-year undergraduates who are aspiring to get a university intern. Senior UG students shared valuable tips and experiences, and the Q&A session facilitated direct interaction. The outcomes include increased awareness, enhanced understanding of application processes, and motivation for pursuing internships at renowned universities.

The speakers of the session were:

1. Utkarsh Gupta (RIYA)
2. Rishi Tunungtula (International visiting internship student program)
3. Manasi Pawar (McKelvey International Students Research Internship Program - Washington University)
4. Isha Dev (ThinkSwiss)

Link to the resources of the session (available through IITB GSuite Account)

[University Intern Session Slides](#)

## ● GRAD PROGRAM

*In coordination with the Career Cell, UGAC*

EnPoWER in collaboration with Career Cell, organized the GRAD Program to aid students who want to pursue further studies in universities abroad and in India. This program consisted of a session, a mentorship program, and a booklet.

### 1. GRAD School 101:

The objective of the GRAD School 101 session was to assist students in navigating the university application process for further studies. In collaboration with Videsh, the session featured three co-founders of the organization as guest speakers, all of whom held prestigious PhDs from renowned universities such as MIT and Purdue. The session encompassed a wide range of topics, including essential components such as crafting a compelling Statement of Purpose (SoP), securing strong Letters of

Recommendation (LoR), initiating the university shortlisting process, and building a robust academic profile.

[GRAD School 101](#)

## 2. GRAD Mentorship Program:

The motivation behind this mentorship program is to help students who want to pursue further research in their topic of interest in global universities. Final year IITB students were allotted mentors belonging to various domains in different countries who have just graduated from IITB and secured admits into top universities around the world. These mentors will help them apply to programs that best align with their research interests.

 GRAD Mentorship 2023 - Mentor Profiles.pdf

 Mentor\_Handbook.pdf

## 3. GRAD Booklet:

The GRAD Booklet aims to provide students with resources to help students with their university application with resources. Insights were collected from various seniors who are pursuing their further studies in top institutions and then compiled in the booklet. Students were also provided with the contact details of seniors and their profiles in different sections of the booklet.

 GRAD Booklet\_ Edition Two.pdf

## ● RESUME MAKING & APPING SERIES

*In coordination with the Career Cell, UGAC*

EnPoWER in association with Career Cell successfully conducted a 3-session series aiming at helping the second-year students in their resume-making and Apping journey.

### 1. Introduction to Resume Making (conducted on 1st October 2023):

This online session was designed to guide sophomores through the fundamental aspects of crafting a resume

The speakers for the session were:

- 1) Aum Samel

2) Shreedar Malpani

2. Formatting the Resume (held on 4th October 2023):

This session, conducted online, focused on aiding second-year students in properly formatting their resumes using tools like Microsoft Word and Overleaf (for LaTeX)

The speakers for the session were:

- 1) Harsh Shah
- 2) Himank Gupta

3. Apping and Univ Programs( held on 26th October 2023):

The concluding session of the series, conducted offline, covered the basics of apping in both core and non-core domains

The speakers for the session were:

- 1) Company Apping: Tashvi Gupta
- 2) Univ Programs: Devashish Bhawe
- 3) Univ Apping: Aaryan Gupta

Link to the resources of the third session:

 [Apping\\_Session.pdf](#)

- **SOPHOMORE 101 SESSIONS (BRANCH INDUCTION SESSIONS)**

*In coordination with Student Support Services, UGAC*

A series of sessions were conducted in 13 UG departments in collaboration with Student Support Services, Department Research Coordinators, and the Student Mentorship Programme team to create awareness among students about the opportunities within their department, the various career options available to them in their core streams and also introducing students to the new curriculum. Additionally, we created a guideline for speakers on how to cover various topics, from research as an undergraduate to core fields in the future. A dashboard was maintained and the sessions were publicised institute-wide to allow students to attend sessions of other departments as well, aiding the decision of branch change for them to be sophomores. A repository of the recordings and slides for the sessions was maintained and released for future access. EnPoWER coordinators also discussed various opportunities for UGs to explore research in their departments.

Link:  [Sophomore 101 Dashboard 2023-24](#)

- **INDUSTRIAL COLLABORATIONS**

EnPoWER collaborates with various industries to bring practical on-ground industrial projects for Undergraduate Students. These included topics like app development marketing, educational content creation, finite element analysis, etc. These were funded industrial projects.

1. **Vizuara:**

Vizuara is a fast-growing US-based EdTech startup. Their founding team consists of graduate students from MIT and Purdue working to make education better. Their singular mission is to deliver the highest quality 3D, gamified, and AR/ VR powered immersive content to schools. They have secured pilot studies with 10 schools globally and demonstrated their product in more than 30+ schools.

The students of IIT Bombay were given the opportunity to the role of project leaders in the following profiles-

- 1) Unity Game Development: Vizuara was looking to add passionate game developers in Unity to convert physics, chemistry, and biology concepts to interactive games.  
Number of students selected - 1

- 2) Science Content Writer: This role involved writing scripts/storyboards for physics/chemistry/biology concepts from Class 5 - Class 10  
Number of students selected - 1

2. **JSW Steel:**

EnPoWER in collaboration with JSW Steel provided an opportunity for IITB students to participate in the steelChallenge-18 competition. steelChallenge-18 is a unique worldwide competition designed to engage students and industry professionals in demonstrating their ability to produce the best quality steel at the lowest cost. This event offers an opportunity for participants to learn, grow, and connect with leaders in the steel industry.

EnPoWER, in collaboration with JSW, gave out 10 free promo codes to interested participants which exempted them from paying the ₹14.49 registration fee for the challenge. Students were selected rigorously based on their resumes and SOP so that the most deserved candidates to ensure the most qualified participants. The competition also offered access to the Learning Solutions Directory, featuring e-learning courses, live masterclasses, and practical simulations.

- **INITIATIVES WITH KCDH**

*In coordination with Student Support Services, UGAC*

The Koita Centre for Digital Health (KCDH) at IIT Bombay was founded in 2021 with the primary goal of driving academic programs, research, and industry collaborations in Digital Health. KCDH addresses the need for engineering, clinical professionals, and entrepreneurs in Digital health and creates research and employment opportunities in this domain. Maintaining the partnership from the previous year and working with the team from KCDH, various initiatives were carried out to create awareness about the field of digital health and inform students about the relatively new IDDDP and minor programs offered by the center.

1. Digital Health Insights - An interactive session was conducted, in collaboration with EnPoWER. The session aimed at providing students with insights on the IDDDP & Minor program provided at the Koita Centre of Digital Health (KCDH) as well as the interesting career possibilities present in this rapidly growing sector. They were also joined by a student of KCDH who shared his experiences.

The panel consisted of:

- Professor Nirmal Punjabi, Adjunct Faculty at KCDH and founder of Sensorama Technologies. Additionally, he is the founder of the Sensing and Monitoring Foundation.
- Professor Ganesh Ramakrishnan, Professor-in-charge at KCDH, Institute Chair Professor at the Department of Computer Science and Engineering, Former Research Staff Member in the IBM India Research Lab

2. Digital Health Talks - Two sessions were conducted with online with the faculty of KCDH and students to discuss the IDDDP Program and Minor in depth.

- The first talk was on the topic “Medical Data-Efficient Machine Learning” and the IDDDP Program. The speakers included Dr. Kshitij Jadhav, MBBS, MD (Pharmacology) from Mumbai's prestigious medical schools (KEM HOSPITAL, and Lokmanya Tilak Municipal General Hospital), Ph.D. in Neuroscience from the University of Lausanne, Switzerland, and a Postdoc at the University of Cambridge, UK in Neuroscience. He was joined by two students who were pursuing an IDDDP from KCDH.

- The second talk was on “Careers in Digital Health” and the Minor Program. One of the speakers was Dr. Richa Singh, Deputy General Manager, Jio HealthHub. She is a doctor



turned digital health expert with 15 years of experience across Public health, Medtech, Pharma, and Digital Health domains with Fortune 500 organizations such as UNICEF, Gates Foundation, CRISIL, Roche, Wipro GE Healthcare. A minor student also shared her experience with the program.

- **SOCIAL MEDIA**

1. **SURP-iSURP'22 Experience Series:**

Experiences of 4 UG researchers in their SURP-iSURP projects were curated in the form of Instagram posts to share advice and insights with the student community

The speakers were as follows:

- 1) Dhananjay Raman
- 2) Tharun Sidambaram
- 3) Soham Joshi
- 4) Nayantara Ramakrishnan

2. **Research Bites:**

A new series featuring the latest updates in the field of research was started this year. So far, 2 posts have been created for the same:

- 1) The first post discussed the experiments to be conducted in the Chandrayaan mission, which focused on the historic landing of Chandrayaan-3
- 2) The second post provided a glimpse into the achievements of 2023 Nobel laureates

3. **Novice's Guide:**

We developed a series of weekly articles to offer a thorough walkthrough of the journey that newcomers in the field of research would typically embark on. This initiative was designed to help students in their pursuit of research knowledge and skills.

4. **Research Spotlight:**

We created a 2 post series on the recipients of the IIT Bombay Research Awards 2022. This series aimed to highlight the contribution of our professors to academia and society. The following professors were featured in our post:

| <i>Professor</i> | <i>Department</i>      | <i>Award</i>                  |
|------------------|------------------------|-------------------------------|
| Prof. Ankit Jain | Mechanical Engineering | Early Research Achiever Award |

|                                  |                                            |                                                      |
|----------------------------------|--------------------------------------------|------------------------------------------------------|
| Prof. Karthikeyan Lanka          | Centre of Studies in Resources Engineering | Early Research Achiever Award                        |
| Prof. Arnab Dutta                | Chemistry                                  | Early Research Achiever Award                        |
| Prof. Rodney A. Fernandes        | Chemistry                                  | Research Dissemination Award                         |
| Prof. Suvarn S Kulkarni          | Chemistry                                  | Research Publication Award                           |
| Prof. Krishna Shankara Narayanan | Computer Science & Engineering             | Research Publication Award                           |
| Prof. Prabhakar Naraga           | Earth Sciences                             | Research Publication Award                           |
| Prof. Swatantra Pratap Singh     | Environmental Science & Engineering        | Prof. Krithi Ramamritham Award For Creative Research |
| Prof. Biswabandan Panda          | Computer Science & Engineering             | Prof. Krithi Ramamritham Award For Creative Research |

## 5. Craft Your Research Persona:

In addition to the session on academic writing, we had a post and a series of stories on our social media on making an academic website:

- 1) The post consisted of a brief insight into the benefits of creating an academic website followed by a 5 step guide on how to create a website.
- 2) In addition to that, we put a series of stories with screenshots of the academic websites of students and alums from IIT Bombay.

## ● INAUGURATION OF RESEARCH PARK (20th February 2024)

The IIT Bombay Research Park- a pivotal center for innovation and collaboration between academia and industry was inaugurated virtually by Shri Hon'ble Prime Minister Narendra Modi, on 20th February 2024. EnPoWER, the only student body was privileged to be invited

to this momentous event, witnessing the unveiling of the new Research Park infrastructure- a 14-floor building with 5 lakhs of built-up space. The event was graced by the esteemed presence of Member of Parliament Shri Manoj Kotak (Mumbai North-East), Chairperson of the Board of Governors of IIT Bombay Shri. Sharad Saraf, Director of IIT Bombay Prof. Subhasis Chaudhuri, Professor-In-Charge of Aspire – IIT Bombay Research Park Prof. Swaroop Ganguly, and various Institute functionaries.

## **Work Done as Institute Secretary of Academic Affairs and as a part of the Council Duties**

- Ensured smooth **Academic Registration** of 4000+ students
- **Freshmen Orientation (31st July 2023 to 4th August 2023)**
  1. Ensured smooth onboarding of first-year UG students and collaborated within and outside of the UGAC to organize the official 5-day-long freshmen orientations
  2. Headed the Venue committee of the Freshmen Orientation to track and book venues for the simultaneous live-streaming of the orientation for 1800+ parents alongside regular lectures for senior UG students and also tracked and booked venues for Orientations on Day 2 and Day 3
  3. Coordinated with 11 UG Departments to help smooth the registration process for students, book venues for department orientations, and guide students from the main convocation hall to the respective department venues
  4. Coordinated with the hostel council and other divisions of the UGAC to organize the Hostel and Academic Orientations for UG freshmen students
  5. Helped in organizing orientations for the Gender Cell, SC/ST Cell, and the SWC
  6. Conducted 1 of the 4 parallel sessions in the Technical Orientation organized by SSS
- **First-Ever Commencement Ceremony (2nd May 2023)**
  1. Coordinated with the Public Relations Office, PGAC, and the SMP Team to organize the logistics of the Commencement of 5000+ to-be graduating students
  2. Ensured a smooth photo-op with the Director for all the students in the Convocation Hall, guaranteeing that the event unfolded without any disruptions
  3. Assisted the Public Relations office in screening the Commencement Ceremony Session with Hon'ble Minister of Education Shri Dharmendra Pradhan's talk in VMCC and LHC rooms

- **Convocation Ceremony (19th and 20th August 2023)**
  1. Coordinated with the Public relations office and the PGAC to organize the logistics of the Convocation of 5000+ graduating students
  2. Headed the team of volunteers from the UGAC to help in the smooth on-ground execution of the Convocation
  
- **Teacher's Day Celebration**
  1. Coordinated with the Public Relations Office to help in the on-ground execution of the Teacher's Day Celebration
  2. Coordinated with the Academic Section to smoothly organize the felicitation of the Institute Academic Prize Winners
  
- Assisted the Academic Section in smooth **Document Verification** of UG freshmen students
  
- **Talk and interaction with the Vice President of India (6th November 2023)**
  1. Coordinated with the Public Relations Office and Dean SA Office to conduct the talk and interaction by the Hon'ble Vice President of India Shri. Jagdeep Dhankhar on 'Role of Youth in Nation Building' as part of Viksit Bharat @ 2047
  2. Prepared and presented a question to the Honorable Vice President as part of the Q&A session with students, centered around the Viksit Bharat initiative
  
- **Dedication of CAR-T Cell Therapy to the Nation by the President of India (4th April 2024)**

Coordinated with the Public Relations Office to ensure the seamless execution of the event, while also coordinating with security to facilitate the addresses by the Honorable President of India, Smt. Droupadi Murmu, and the Honorable Governor of Maharashtra, Shri. Ramesh Bais, focuses on 'CAR-T Cell Therapy' as a domestically developed and cost-effective treatment option for blood cancer patients in the nation.